PROGRESS REPORT

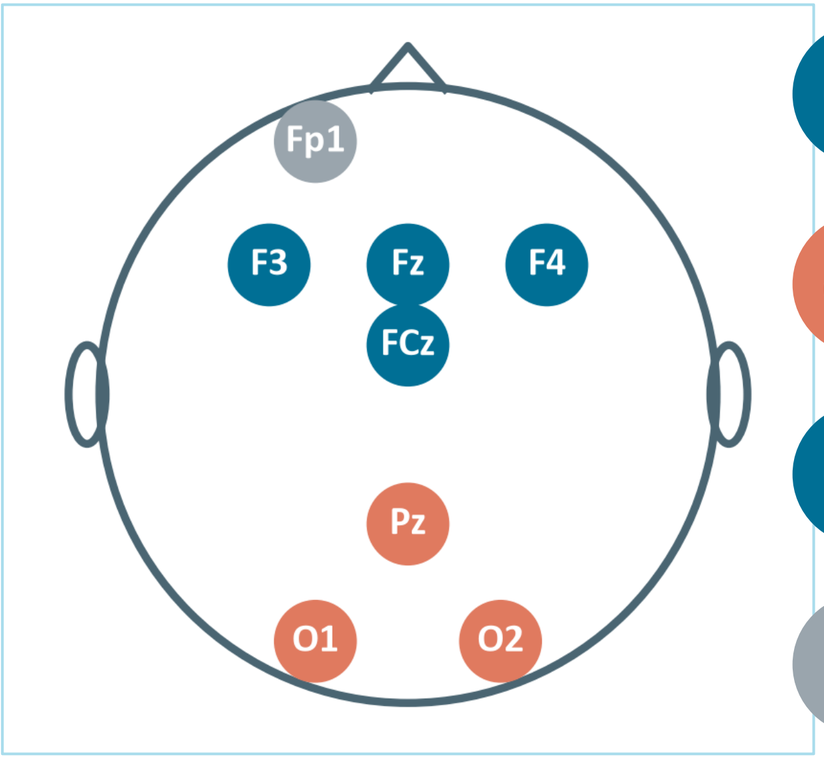
Closed-Loop Neuroadaptive AR Platform for **Scrolling**



RICARDO MONRREAL POOL

Scope & Framework

We measure the **cognitive cost** of scrolling



θ ↑ **Tonic theta → Fatigue**
Fz · FCz · F3 · F4

α ↑ **Periodic alpha → Disengagement**
Pz · O1 · O2

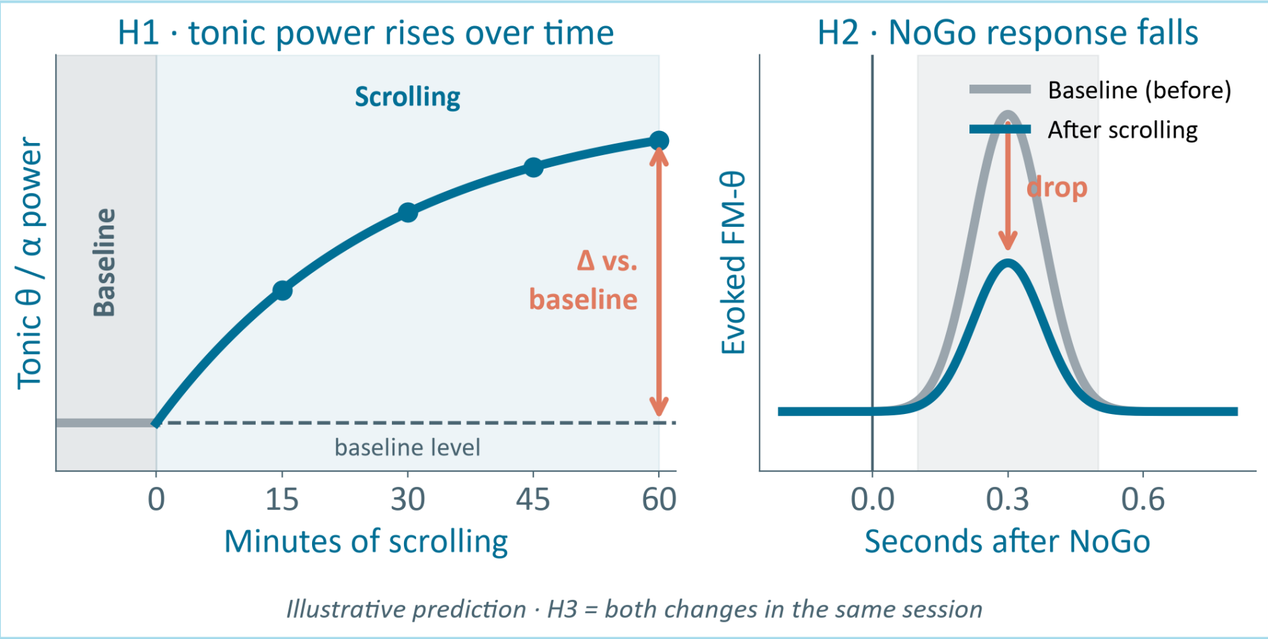
θ ↓ **NoGo theta → Loss of inhibitory control**
Fz · FCz · phasic, after NoGo

Fp1 **Blinks → Artifact veto**
Fp1

⚠ **Out of scope: dopamine** · VTA / striatum are unreachable by scalp EEG

Hypotheses

The original **time-based** model



Illustrative prediction · H3 = both changes in the same session

H1 **Tonic θ / α ↑ more with scrolling than with a documentary**

Literature ✓ · datasets partial

H2 **NoGo FM-θ ↓ and commission errors ↑ after scrolling**

Pending · own Go/NoGo

H3 **Both at once: tonic ↑ / phasic ↓**

Precedent: Arnau 2021

Source: Arnau, Brümmer, Liegel & Wascher (2021), Psychophysiology, 58(6), e13805

Theoretical Grounding

H1

g = 1.03

Theta ↑ with fatigue

Tran 2020 · meta-analysis

H1

g = 0.85

Alpha ↑ with fatigue

Tran 2020 · meta-analysis

H2

NoGo errors ↑

45 min smartphone vs. documentary

Jacquet 2023 · behavior, no EEG

~1 h

Alpha peaks, theta keeps rising

Wascher 2014

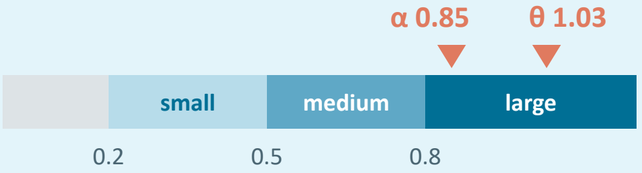
H2 H3

Tonic ↑ · Phasic ↓

Direct precedent of H3

Arnau 2021 · Yu 2022

Effect size scale (Hedges' g)



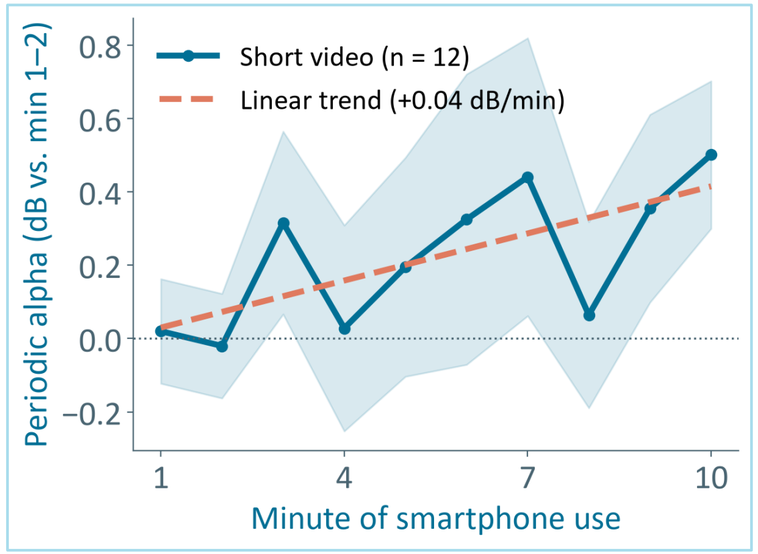
Both effects are large (≥ 0.8)

Requires **≥ 30–60 min** · demanding task · **never tested with EEG + NoGo during short-video scrolling**

Sources: Tran et al. (2020), Psychophysiology · Wascher et al. (2014), Biol. Psychol. · Arnau et al. (2021), Psychophysiology · Yu et al. (2022), Sci. Rep. · Jacquet et al. (2023), Sci. Rep.

Practical Validation

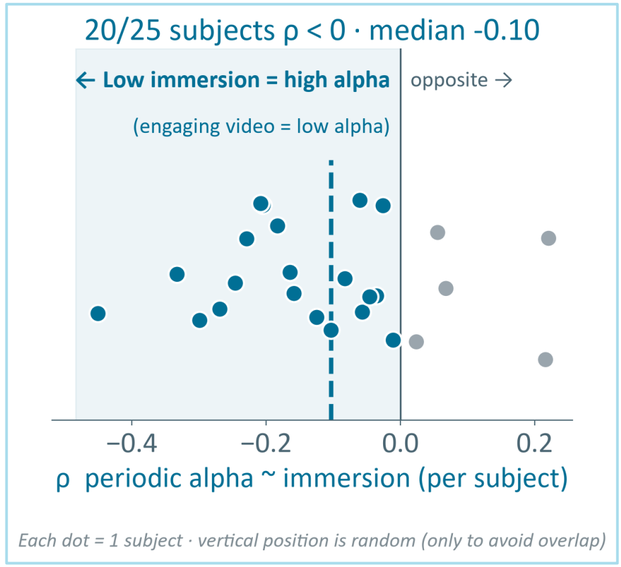
ds007537 · smartphone · 10 min



Alpha ↑ +0.43 dB

Short video · 10/12 subjects

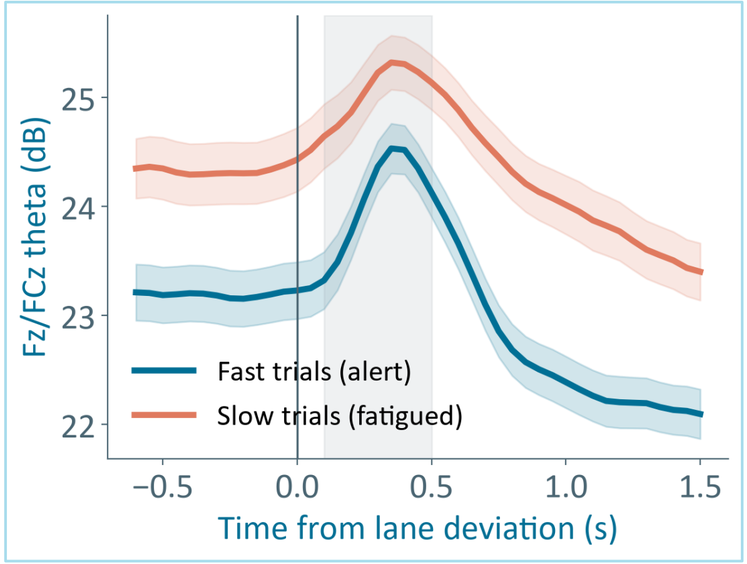
EEG-SVRec · short video · ≤ 30 min



Immersion ↑ → alpha ↓

20/25 subjects

Cao 2019 · driving · 60–90 min



Tonic θ ↑ → slow RT

27/27 · phasic preserved

Sources: Mishra et al. (2026), OpenNeuro ds007537 · Zhang et al. (2024), EEG-SVRec, SIGIR '24 · Cao et al. (2019), Scientific Data, 6, 19

Why the Datasets Could Not Confirm It

	ds007537	EEG-SVRec	Cao 2019	Our study
Duration ≥ 30 min	✗ 10 min	✗ ≤ 30 min	✓ 60–90 min	✓ 45–60 min
NoGo probe	✗ free use	✗ passive	✗ lane deviation	✓ in-feed + pre/post
Control condition	✗ fixed order	✗ fixed order	✗ none	✓ documentary
Fatigue anchor	✗ none	✗ immersion only	✓ RT, lapses	✓ KSS + RT
Scrolling paradigm	✓ smartphone	✓ short video	✗ driving	✓ short video

To our knowledge, no public dataset has all five

Sources: Mishra et al. (2026), OpenNeuro ds007537 · Zhang et al. (2024), EEG-SVRec, SIGIR '24 · Cao et al. (2019), Scientific Data, 6, 19

Experimental Plan

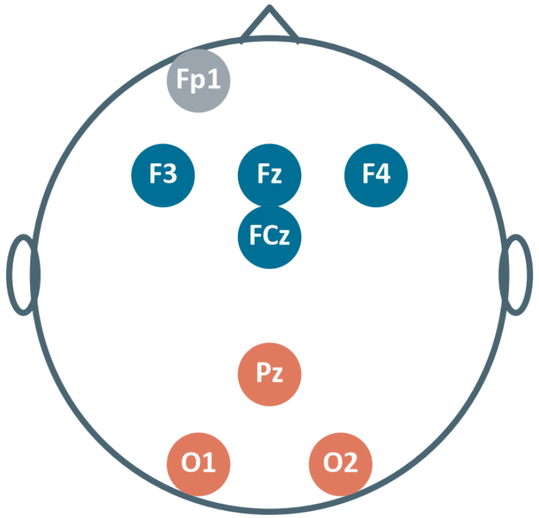
Baseline
1 min

Go/NoGo
pre · 3 min

Scrolling | documentary
45–60 min · in-feed NoGo

Go/NoGo
post · 3 min

Self-report
KSS · RT



AURA · 8 channels

Frontal theta
Fz · FCz · F3 · F4

Posterior alpha
Pz · O1 · O2

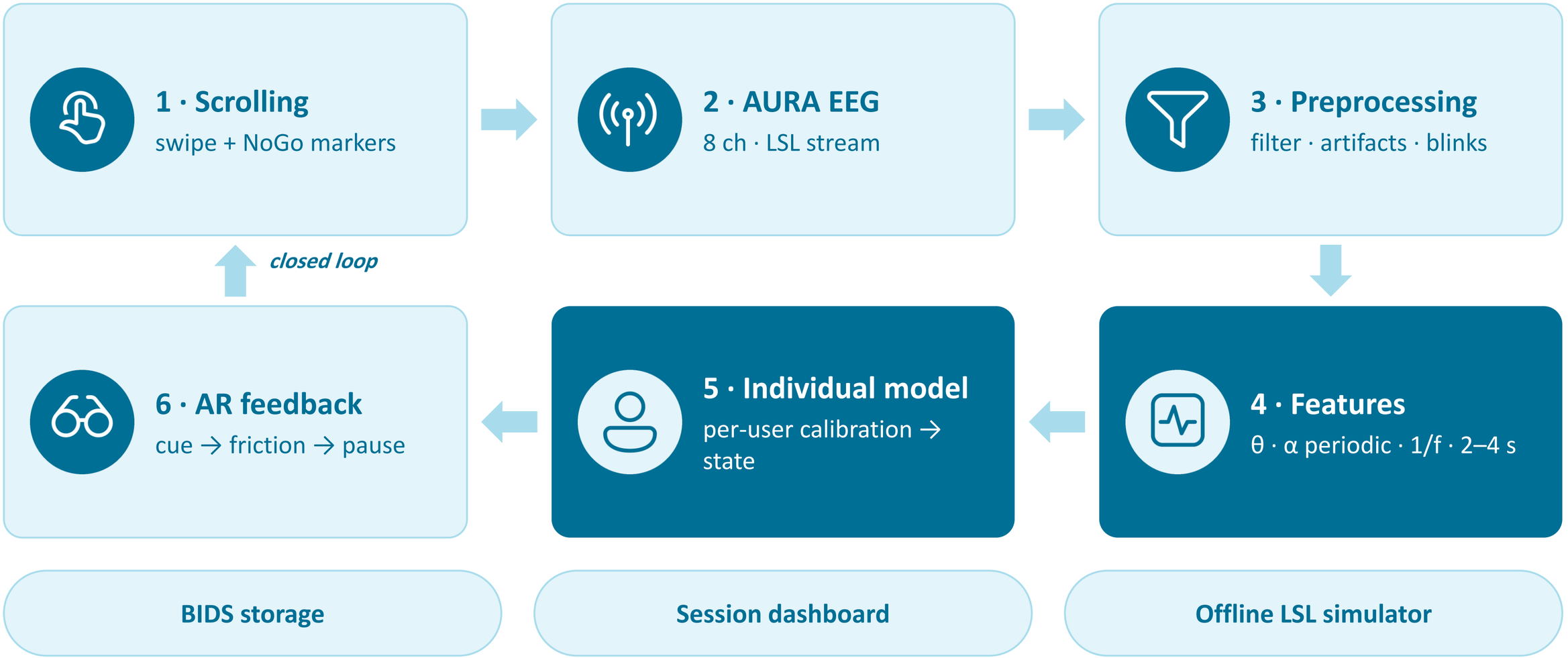
Blinks
Fp1

Simulation first

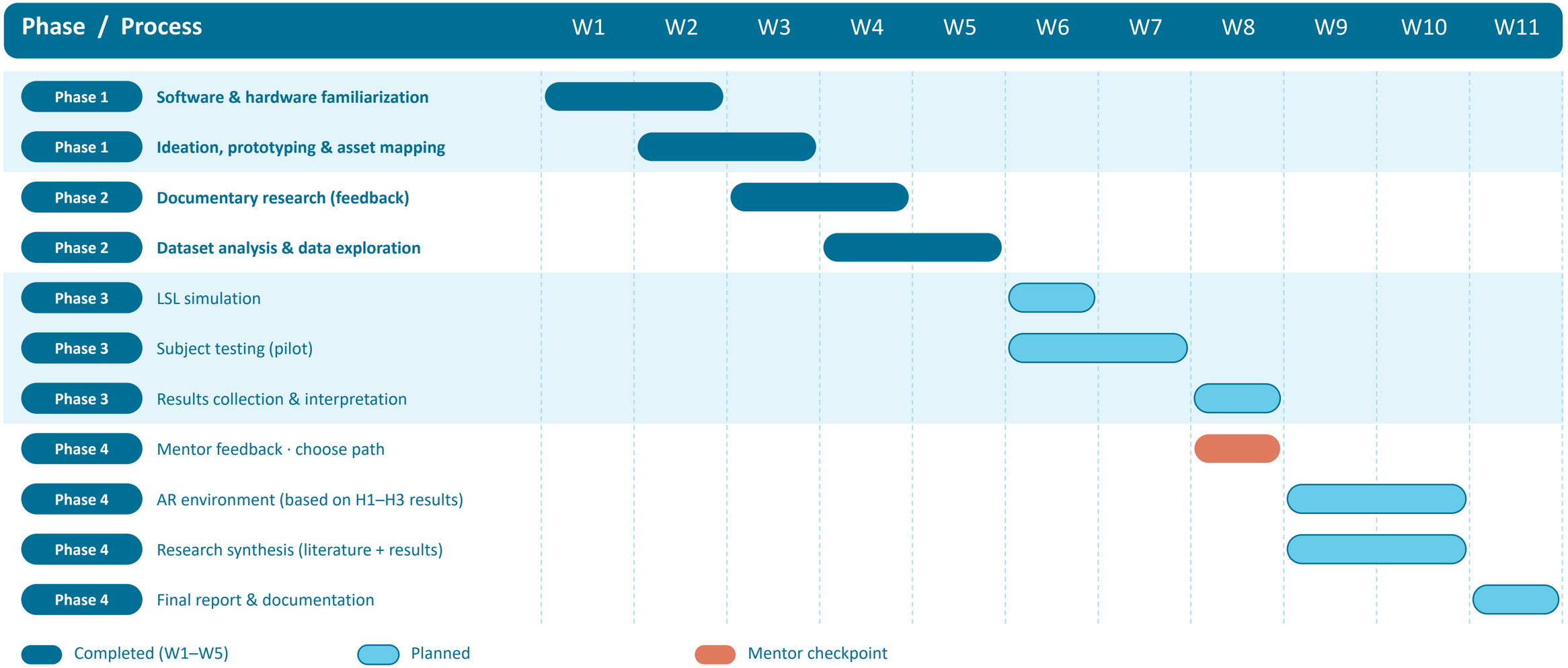
- LSL replay of recorded EEG
- Tests features + latency

EEG file → LSL replay → Pipeline → AR loop

System Pipeline



Gantt Diagram



Next Steps

1. LSL simulation with the public datasets
2. Own experiment: EEG data from subjects
3. Validate H1, H2 and H3

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